

Climate 5000 Series - Gen 4

18K BTU Single Zone Max Performance System

1-Way Cassette IDU

Engineering
Submittal
Sheet



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Overview and Project Information



Project name

Location

Architect

Engineer

General contractor

Reference #

18K Btu Single Zone Max Performance System - 1-Way Cassette IDU

Indoor part number	8-733-962-668
Indoor model number	BMS500-AAU018-1AHZXD
Outdoor part number	8-733-962-690
Outdoor model number	BMS500-AAS018-1CSXHD

Efficiency Data		AHRI 210/240 2023
SEER2		20.6
EER2		12.5
HSPF2 Region IV		12.2
HSPF2 Region V		9.2
COP(47F)		3.01
COP (5F)		1.89

Performance Data		AHRI 210/240 2023
Cooling	Capacity Range	5700-19000
	Rated Capacity	16700
	Current	8.3
	Input Power	1337
Heating	Capacity Range	8900-22000
	Rated Capacity (47F)	20000
	Rated Capacity (5F)	17500
	Current	12.0
	Input Power	1950

Electrical Data	
Power supply (V / Ø / Hz)	208-230 / 1 / 60
Min. circuit ampacity indoor/outdoor (A)	3.0 / 19.0
Max. fuse indoor/outdoor (A)	15.0 / 20.0

Compressor Data	
Type:	Rotary
Refrigerant oil /oil charge (ml)	VG74 / 620
Input (W)	805 / 2045
Rated current (RLA/A)	4.15 / 9.30

Piping	
Refrigerant precharge (oz/kg)	55.38 (1.00)
Length for precharge (ft/m)	24.6 / 7.5
Liquid side (in/mm)	1/4" / Ø6.35
Gas side (in/mm)	1/2" / Ø12.7
Additional Charge of Refrigerant (oz/ft)/(g/m)	0.16 / 15
Max. refrigerant pipe length (ft/m)	98.43 / 30
Max. difference in height (ft/m)	65.62 / 20

Operation Range

ODU	Cooling (°F / °C)	-22~122 / -30~50
	Heating (°F / °C)	-22~75 / -30~24
IDU	Cooling (°F / °C)	60~90 / 16~32
	Heating (°F / °C)	32~86 / 0~30

System Data

Refrigerant type		R454B
Design pressure (PSIG)		550 / 340
Dehumidification (pts/hr)		4.33
IDU	Fan motor model	ZKFN-30-8-43
	Fan motor qty (ea)	1
	Fan motor input (W)	46.0
	Fan speed (RPM)	1344 / 1184 / 1024
	Air flow, Turbo / H / M / L / Si (CFM)	400.2 / 352.0 / 300.2 / 207.2 / N/A
	Approx. Air Throw Range ft (m)	24.6 (7.5)
	Noise level, Turbo / H / M / L / Si (dB(A))	N/A / 44 / 41.5 / 31 / 26
	Net weight (lbs / kg)	45.19 / 20.5
ODU	Gross weight (lbs / kg)	82.89 / 37.6
	Drain pipe diameter (mm)	ODØ25mm
	Fan motor model	ZKFN-80-8-3
	Fan motor qty (ea)	1
	Fan motor input (W)	115
	Fan motor speed (RPM)	780/700/550
	Air flow, Max (CFM)	1765.8
	Noise level, Max (dB(A))	59.0
	Net weight (lbs / kg)	99.87 / 45.3
	Gross weight (lbs / kg)	107.8 / 48.9

Accessories

BMS-WT2-XXD (8733962707)	Wired Wall Thermostat
BMS-DRY-XXD (8733969263)	24V Interface
BMS-PK2-7X1 (8733951011)	25ft Lineset

Standard Parts

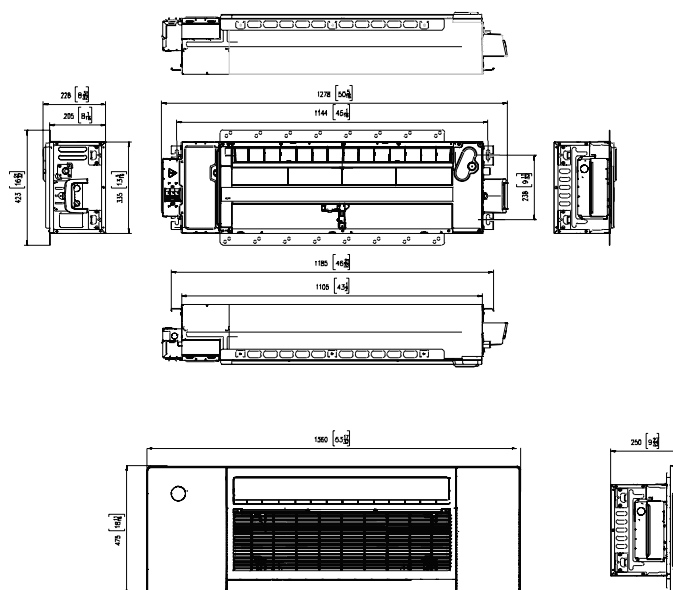
8-733-969-065	Remote Control
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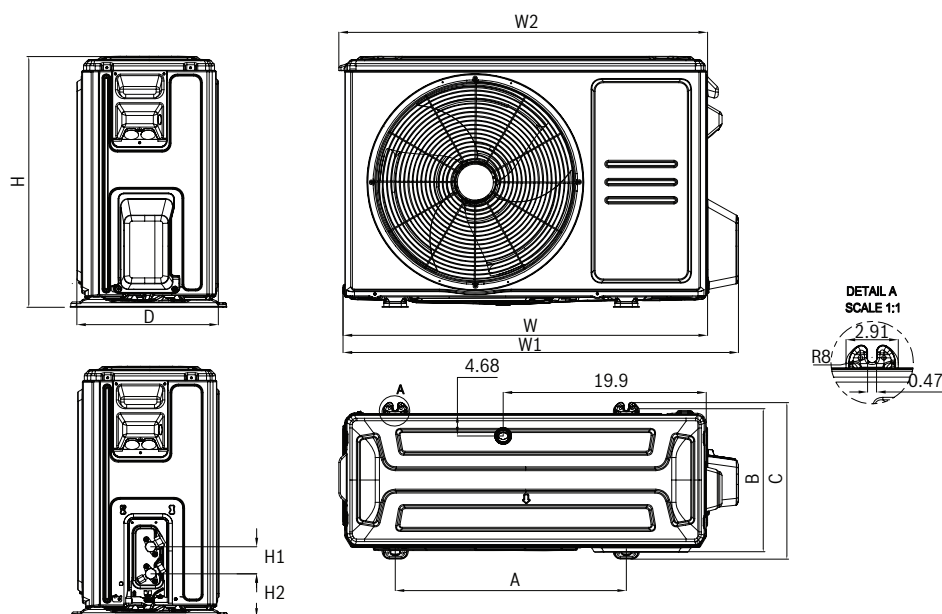
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Indoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)
BMS500-AAU018-1AHZXD	50.31	13.19	8.98

Outdoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)	W1 (in)	W2 (in)	A (in)	B (in)	C (in)	H1 (in)	H2 (in)
BMS500-AAS018-1CSXHD	35	13.46	26.5	37.6	35.24	26.1	13.7	14.96	2.36	4.25

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Cooling Performance - Indoor Dry Bulb Temperature at 400 CFM																					
ODU DB(°F)	IDU WB (°F)	60.8°F(16.0°C)					64.4°F(18.0°C)					67°F(19.4°C)					71.6°F(22.0°C)				
	IDU DB (°F)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)
-22°F (-30.0°C)	TC	14027	14021	14221	14280	14673	14636	14624	14614	14613	14601	15073	15054	15040	15038	15024	15899	15895	15862	15856	15823
	S/T	0.80	0.91	0.99	0.99	0.99	0.64	0.75	0.84	0.86	0.97	0.55	0.65	0.74	0.75	0.86	0.39	0.47	0.55	0.57	0.67
	PI	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76
-13°F (-25.0°C)	TC	15196	15189	15406	15470	15896	15855	15842	15832	15831	15818	16329	16309	16293	16291	16275	17224	17220	17184	17178	17141
	S/T	0.80	0.91	1.00	1.00	1.00	0.65	0.75	0.85	0.86	0.97	0.55	0.65	0.74	0.76	0.86	0.39	0.47	0.56	0.57	0.67
	PI	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79
-4°F (-20.0°C)	TC	16365	16358	16591	16660	17119	17075	17061	17050	17048	17034	17585	17563	17547	17544	17527	18549	18544	18506	18499	18460
	S/T	0.80	0.92	1.00	1.00	1.00	0.65	0.76	0.85	0.87	0.98	0.55	0.65	0.74	0.76	0.86	0.39	0.48	0.56	0.57	0.67
	PI	0.82	0.82	0.82	0.82	0.82	0.82	0.82	0.82	0.82	0.82	0.81	0.82	0.82	0.82	0.82	0.82	0.82	0.82	0.82	0.82
0°F (-17.8°C)	TC	16909	16899	17212	17284	17759	17634	17614	17601	17598	17775	18152	18126	18104	18101	18092	19128	19124	19122	19125	19137
	S/T	0.81	0.93	1.00	1.00	1.00	0.66	0.77	0.86	0.88	1.00	0.55	0.66	0.75	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.84	0.84	0.84	0.84	0.84
5°F (-15.0°C)	TC	17393	17453	17852	17934	18487	18209	18214	18215	18234	18505	18791	18796	18800	18800	18825	19894	19887	19898	19900	19909
	S/T	0.82	0.94	1.00	1.00	1.00	0.66	0.78	0.87	0.88	1.00	0.56	0.67	0.76	0.78	0.88	0.39	0.48	0.57	0.59	0.69
	PI	0.83	0.83	0.84	0.84	0.84	0.84	0.84	0.84	0.84	0.84	0.84	0.84	0.84	0.84	0.84	0.86	0.86	0.86	0.86	0.86
14°F (-10.0°C)	TC	18257	18248	18620	18708	19288	19163	19147	19134	19131	19311	19820	19795	19775	19772	19752	21075	21068	21028	21021	20976
	S/T	0.81	0.93	1.00	1.00	1.00	0.65	0.76	0.86	0.87	1.00	0.55	0.66	0.75	0.76	0.87	0.39	0.48	0.56	0.57	0.67
	PI	0.67	0.67	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.67	0.67	0.67	0.67	0.67
17°F (-8.3°C)	TC	18501	18561	18990	19083	19703	19422	19430	19433	19450	19724	20085	20092	20100	20101	20126	21360	21355	21351	21355	21373
	S/T	0.82	0.93	1.00	1.00	1.00	0.66	0.77	0.87	0.88	1.00	0.55	0.66	0.76	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.61	0.61	0.61	0.61	0.61
23°F (-5.0°C)	TC	18676	18756	19241	19336	19974	19622	19627	19630	19655	19995	20298	20305	20310	20311	20344	21583	21576	21592	21595	21607
	S/T	0.83	0.95	1.00	1.00	1.00	0.66	0.78	0.88	0.89	1.00	0.56	0.67	0.76	0.78	0.89	0.39	0.48	0.57	0.59	0.69
	PI	0.65	0.65	0.65	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.67	0.67	0.67	0.67	0.67
35°F (1.7°C)	TC	19205	19205	19318	19372	19997	20102	20102	20103	20103	20157	20732	20730	20731	20732	20734	21899	21894	21893	21892	21894
	S/T	0.78	0.88	0.97	0.99	1.00	0.63	0.73	0.82	0.83	0.93	0.54	0.63	0.72	0.73	0.83	0.39	0.46	0.54	0.56	0.65
	PI	0.70	0.70	0.71	0.71	0.74	0.75	0.75	0.75	0.75	0.75	0.78	0.78	0.78	0.78	0.78	0.85	0.85	0.85	0.85	0.85
41°F (5.0°C)	TC	18980	18980	19131	19264	19834	19863	19862	19863	19863	19933	20482	20481	20482	20483	20484	21632	21626	21625	21624	21626
	S/T	0.78	0.89	0.98	1.00	1.00	0.64	0.74	0.83	0.84	0.94	0.54	0.64	0.72	0.74	0.84	0.39	0.47	0.55	0.56	0.66
	PI	0.76	0.76	0.77	0.77	0.80	0.81	0.81	0.81	0.81	0.81	0.84	0.84	0.84	0.84	0.84	0.92	0.92	0.92	0.92	0.92
47°F (8.3°C)	TC	18755	18767	19019	19104	19669	19624	19623	19624	19624	19715	20234	20232	20233	20233	20233	21366	21361	21359	21358	21360
	S/T	0.79	0.90	1.00	1.00	1.00	0.64	0.75	0.84	0.85	0.96	0.54	0.65	0.73	0.75	0.85	0.39	0.47	0.55	0.57	0.67
	PI	0.81	0.81	0.83	0.83	0.87	0.87	0.87	0.87	0.87	0.87	0.91	0.91	0.91	0.91	0.91	0.99	0.99	0.99	0.99	0.99
50°F (10.0°C)	TC	18423	18930	19569	19682	20445	19463	19474	19608	19670	20465	20222	20249	20259	20260	20455	21616	21673	21729	21735	21763
	S/T	0.90	1.00	1.00	1.00	1.00	0.71	0.85	0.96	0.99	1.00	0.59	0.72	0.83	0.85	0.99	0.40	0.50	0.60	0.62	0.74
	PI	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93
59°F (15.0°C)	TC	18435	19082	19759	19879	20688	19505	19532	19753	19901	20709	20310	20330	20337	20367	20726	21809	21852	21896	21901	21919
	S/T	0.92	1.00	1.00	1.00	1.00	0.73	0.86	1.00	1.00	1.00	0.60	0.73	0.85	0.86	1.00	0.40	0.51	0.61	0.63	0.76
	PI	1.00	1.00	1.01	1.01	1.01	1.01	1.01	1.01	1.01	1.01	1.01	1.01	1.01	1.01	1.01	1.02	1.02	1.02	1.02	1.02
68°F (20.0°C)	TC	18438	19225	19938	20065	20916	19520	19564	19960	20086	20937	20363	20379	20415	20433	20954	21942	21982	22017	22020	22032
	S/T	0.95	1.00	1.00	1.00	1.00	0.74	0.89	1.00	1.00	1.00	0.61	0.75	0.87	0.89	1.00	0.40	0.51	0.63	0.65	0.78
	PI	1.07	1.08	1.08	1.08	1.09	1.08	1.08	1.08	1.08	1.09	1.09	1.09	1.09	1.09	1.09	1.10	1.10	1.10	1.10	1.10
77°F (25.0°C)	TC	18426	19313	20069	20203	21105	19534	19596	20091	20225	21126	20418	20441	20493	20516	21144	22053	22120	22171	22176	22197
	S/T	0.97	1.00	1.00	1.00	1.00	0.75	0.90	1.00	1.00	1.00	0.61	0.76	0.88	0.90	1.00	0.40	0.52	0.63	0.65	0.79
	PI	1.15	1.16	1.16	1.16	1.17	1.16	1.16	1.16	1.16	1.17	1.16	1.16	1.16	1.16	1.17	1.18	1.18	1.18	1.18	1.17
82°F (27.8°C)	TC	18246	19177	19951	20086	21006	19335	19409	19968	20108	21028	20234	20260	20323	20348	21046	21877	21960	22017	22023	22047
	S/T	0.99	1.00	1.00	1.00	1.00	0.76	0.91	1.00	1.00	1.00	0.62	0.77	0.89	0.92	1.00	0.40	0.52	0.64	0.66	0.80
	PI	1.24	1.25	1.25	1.25	1.26	1.25	1.25	1.25	1.25	1.26	1.26	1.26	1.26	1.26	1.26	1.27	1.27	1.27	1.27	1.27
86°F (30.0°C)	TC	17944	18840	19610	19747	20667	18923	19010	19632	19770	20688	19820	19843	19916	19944	20706	21458	21540	21592	21597	21617
	S/T	1.00	1.00	1.00	1.00	1.00	0.77	0.93	1.00	1.00	1.00	0.62	0.78	0.91	0.93	1.00	0.40	0.52	0.64	0.66	0.81
	PI	1.38	1.38	1.39	1.39	1.40	1.38	1.38	1.39	1.39	1.40	1.39	1.39	1.39	1.39	1.40	1.41	1.41	1.41	1.41	1.41
95°F (35.0°C)	TC	17128	17220	17777	17882	18589	18203	18214	18220	18241											

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Heating Performance - Indoor Dry Bulb Temperature at 400 CFM										
Outdoor DB(°F)	TC Total Heating Capacity(BTU/hr)					PI Power Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-22°F(-30.0°C)	9743	9548	9500	9462	9377	1.90	1.92	1.92	1.92	1.93
-13°F(-25.0°C)	11281	11055	11000	10956	10857	2.06	2.08	2.08	2.08	2.09
-4°F(-20.0°C)	14870	14573	14500	14442	14312	2.27	2.29	2.30	2.30	2.31
0°F(-17.8°C)	16238	15913	15833	15770	15628	2.46	2.48	2.48	2.49	2.50
5°F(-15.0°C)	17947	17588	17500	17430	17273	2.69	2.71	2.71	2.72	2.73
14°F(-10.0°C)	18947	18568	18475	18401	18235	2.67	2.69	2.69	2.70	2.71
17°F(-8.3°C)	19280	18894	18800	18725	18556	2.66	2.68	2.69	2.69	2.70
22°F(-5.6°C)	19530	19139	19044	18967	18797	2.50	2.52	2.53	2.53	2.54
27°F(-2.8°C)	19780	19384	19287	19210	19037	2.34	2.36	2.36	2.37	2.38
32°F(0.0°C)	20030	19629	19531	19453	19278	2.18	2.20	2.20	2.21	2.22
35°F(1.7°C)	20179	19776	19677	19598	19422	2.08	2.10	2.10	2.11	2.12
37°F(2.8°C)	20611	20198	20098	20017	19837	2.06	2.08	2.09	2.09	2.10
42°F(5.6°C)	21689	21255	21149	21064	20875	2.02	2.03	2.04	2.04	2.05
44.6°F(7.0°C)	22120	21678	21569	21483	21290	2.00	2.01	2.02	2.02	2.03
47°F(8.3°C)	22767	22312	22200	22111	21912	1.97	1.99	1.99	1.99	2.00
52°F(11.1°C)	21710	21276	21170	21085	20895	1.72	1.73	1.74	1.74	1.75
57°F(13.9°C)	20654	20240	20139	20059	19878	1.47	1.48	1.48	1.49	1.49
59°F(15.0°C)	20231	19826	19727	19648	19471	1.37	1.38	1.38	1.39	1.39
62°F(16.7°C)	19597	19205	19109	19032	18861	1.22	1.23	1.23	1.23	1.24
64.4°F(18.0°C)	19090	18708	18614	18540	18373	1.10	1.11	1.11	1.11	1.12
69.8°F(21.0°C)	17948	17589	17502	17432	17275	0.83	0.83	0.84	0.84	0.84
75.2°F(24.0°C)	16807	16471	16389	16323	16176	0.56	0.56	0.56	0.56	0.57

Bolded values were tested. All other values are simulated and for reference only.